

Apply a heparin-coated polymer layer to the catheter lumen and integrate a thin piezoelectric strip that emits low-frequency ultrasonic waves (acoustic streaming) along the inner wall to synergistically inhibit clot adhesion.

Executive Technology Brief

Domain: acoustic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'catheter flow occlusion' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 100000: Injury [VITAL SIGNS]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: 'Commentary on: "clinical and haemodynamic evaluation of a novel physician modified inner branch iliac branch device for East Asians".'; 'Biosurfactants as Antibiofilm Agents for Medical Devices: Mechanisms, Evidence and Integration into Infection Prevention and Control.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Anticoagulant heparin coating on the catheter surface reduces thrombus formation, leading to longer failure-free survival.

Architecture: PROPOSED architecture (no physical system exists): transducer element + matching layer; excitation / receive electronics; signal chain (detection criterion); coupling path to target medium — realizing: Apply a heparin-coated polymer layer to the catheter lumen and integrate a thin piezoelectric strip that emits low-frequency ultrasonic waves (acoustic streaming) along the inner wall to synergistically inhibit clot adhesion.

Governing Model: Acoustics / ultrasonics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

AC-002: Wave speed relation. Expression: $c = f * \lambda$. Applicability to this invention: APPLICABLE; holds under: non-dispersive propagation in the medium of interest. Assumptions: sound speed is medium- and temperature-dependent — source the value for the actual medium. Engaged variables: f .

AC-003: Exponential attenuation. Expression: $A = A_0 * \exp(-\alpha * d)$. Applicability to this invention: CONDITIONAL; holds under: homogeneous lossy medium; single frequency. Assumptions: α is frequency- and medium-dependent; two-way path doubles the exponent. Engaged variables: NOT ESTABLISHED.

AC-001: Acoustic impedance contrast (pulse-echo). Expression: $R = ((Z_2 - Z_1) / (Z_2 + Z_1))^2$. Applicability to this invention: CONDITIONAL; holds under: normal incidence, plane wave, lossless media. Assumptions: oblique incidence and attenuation modify detectability. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.